In conclusion, let us discuss a few examples which have played an especially important part in the progress of nuclear physics. Let us take these in their chronological order.

In 1919, Rutherford accomplished the first artificial nuclear transmutation, changing nitrogen atoms into oxygen atoms by bombardment by alpha particles. A proton (being a hydrogen nucleus, it is written $_1H^1$) was liberated in this process, which can be expressed as follows:

$$_7N^{14} + _2He^4 \rightarrow _8O^{17} + _1H^1$$

The oxygen atom thus formed is the rare oxygen isotope of mass number 17. Figure 10 shows such a process in the cloud chamber.

Another important transmutation reaction led to the discovery of the neutron by Joliot-Curie and Chadwick, in 1932. A beryllium nucleus, $_4Be^9$, was bombarded by an alpha particle, and a carbon nucleus, $_6C^{12}$, plus a neutron was obtained, thus:

$$_4Be^9 + _2He^4 \rightarrow _6C^{12} + _0n^1$$

As the third of these important reactions, let us mention the first transmutation of nuclei by artificially accelerated particles, accomplished by Cockroft and Walton, using protons, in the same year, 1932. The protons were accelerated by a high-tension apparatus of 600,000 volts. When such a proton hits a lithium nucleus, $_3Li^7$, the following reaction occurs:

$$_3Li^7 + _1H^1 \rightarrow _2He^4 + _2He^4$$